

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **XCTest-based automated test suites using Swift** for iOS/macOS applications, achieving **92% code coverage** across critical user workflows; integrated test frameworks with **Jenkins CI/CD pipelines** for continuous deployment and real-time test reporting.
- Implemented **LLM-powered test generation and intelligent defect analysis systems** using **GenAI and Large Language Models** to automatically generate test cases from feature specifications, reducing manual test design effort by **45%** and accelerating release cycles.
- Executed comprehensive automated testing across all Apple platforms with detailed **performance metrics and analysis**; designed test strategy covering **localization and internationalization testing** for multi-region deployments and **accessibility testing** compliance validation.
- Developed machine learning model training and fine-tuning pipelines using **Python and scikit-learn** to classify and prioritize defects from automated test outputs, improving defect triage accuracy by **38%** and enabling data-driven release decisions.
- Built REST API integration layers and test orchestration frameworks connecting **XCTest automation suites** with downstream CI/CD systems; leveraged **Git branching, code reviews, and Jira ticketing** for traceability and collaborative quality assurance across distributed teams.
- Collaborated with cross-functional engineering, product, and localization teams to gather test requirements, execute UAT cycles, and provide post-release quality assurance; communicated automation progress and debugging insights through clear documentation and stakeholder presentations.

SKILLS

Automation Testing & Frameworks – XCTest, Swift, automated test suite design, test strategy development, performance metrics analysis.

GenAI & Machine Learning – Large Language Models (LLMs), GenAI for test generation, machine learning model training, scikit-learn, defect classification and analysis.

iOS/macOS Development – Swift programming, iOS/macOS application testing, platform-specific automation, accessibility testing, localization and internationalization testing.

CI/CD & DevOps – Jenkins pipelines, Git branching, continuous deployment, test automation integration, code review workflows.

Data Analysis & Reporting – Python, pandas, NumPy, test metrics dashboards, performance analysis, Power BI reporting.

Software Development – REST APIs, system integrations, Spring Boot, SQL (PostgreSQL, MySQL, Oracle), Docker containerization.

Professional Skills – Problem-solving, debugging, cross-functional collaboration, requirements gathering, UAT coordination, technical communication.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

LLM-Driven XCTest Automation Platform for Multi-Platform Apple Software (HCL Technologies)

- Architected and deployed an **LLM-powered test generation engine** that ingests feature specifications and automatically generates comprehensive **Swift XCTest suites** for iOS, macOS, watchOS, and tvOS; reduced manual test case authoring time by **50%** and increased test coverage from **68% to 94%** across critical user journeys.
- Integrated **machine learning model training pipelines** to analyze test failure patterns and classify defects by severity and component; trained and fine-tuned classification models achieving **89% precision**, enabling automated defect routing and accelerating time-to-resolution by **35%**.
- Implemented **automated localization and accessibility testing workflows** covering **22 languages and WCAG 2.1 compliance standards**; designed test strategies for multi-region deployment validation, reducing localization-related production incidents by **72%** across international releases.
- Connected automation framework to **Jenkins CI/CD pipeline** with real-time test execution, detailed **performance metrics dashboards**, and post-run analysis; published comprehensive test reports via **Power BI** for stakeholder visibility, enabling daily release readiness assessments.
- Collaborated with engineering, QA, product, and localization teams across 3 time zones; documented test strategy, automation architecture, and defect analysis algorithms in Confluence; trained 12+ engineers on LLM-powered test generation methodology and XCTest best practices.

SWIFT Messaging Production Test Automation & Compliance Validation Dashboard (HCL Technologies)

- Designed and executed automated test suites for **SWIFT messaging payment operations** platform supporting **MT/MX message formats, SEPA, cross-border payments, and ISO 20022 compliance**; implemented **performance monitoring and SLA validation tests** ensuring **99.98% transaction accuracy** in production.
- Built **Python automation scripts** for data validation, reconciliation workflow testing, and regulatory compliance checks (**GDPR, PSD2, Open Banking** standards); integrated tests into **Jenkins CI/CD pipelines** with automated post-release verification.
- Created operational dashboards using **Power BI and DAX** to visualize test execution metrics, payment message throughput, and compliance audit trails; presented automated testing results to banking-domain stakeholders and regulatory teams.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)